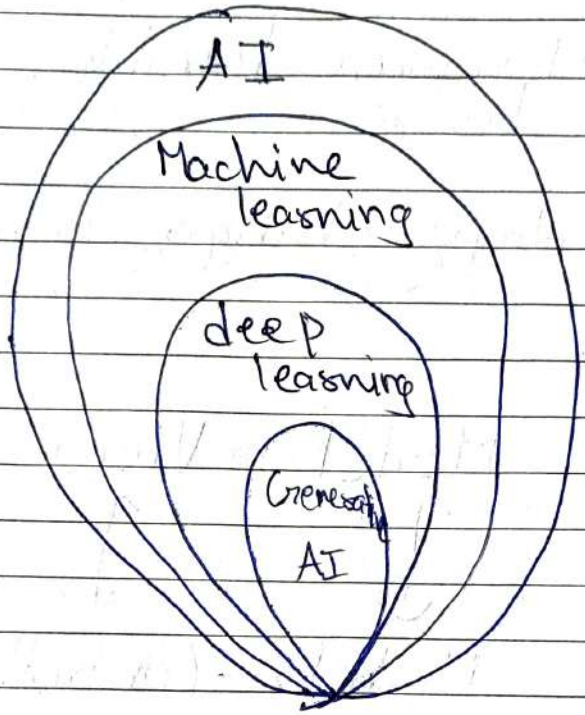
Date _/_/_

AI => Artifical Intelligence is a field of CS dedicated to solving problems that we commonly associate with human intelligence



IT terminology

- Network: cables, routers, and servers connected with each other.
- Routes: A networking device that forwards data packets b/w computer networks. They know where to send packets on the internet.
- Switch: Takes a packet and send it to the correct server/client on your network

Date _/_/_

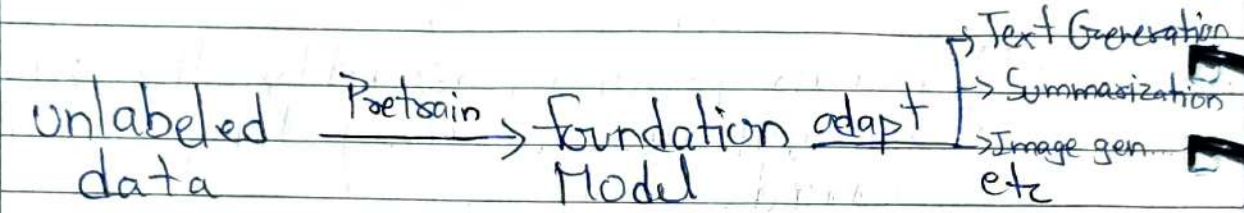
Q What is cloud Computing?

- Cloud computing is the on-demand delivery of compute power, db storage, applications, other IT resources.
- Through a cloud services platform with pay-as-you-go pricing
- You can provision exactly the right type and size of computing resources you need.
- You can access as many resources as you need, almost instantly.
- Simple way to access servers, storage, db and set of application services
- AWS owns and maintains the network-connected hardware required for these application services, while you provision and use what you need via a web application.

Date _/_/_

Q What is Generative AI?

- Generative AI (Gen-AI) is a subset of deep learning.
- Used to generate new data that is similar to data it was trained on
  - Text, image, Audio, etc



Foundational Model

- To generate data, we must rely on a foundational Model
- Foundational Models are trained on a wide variety of input data
- The models may cost tens of millions of dollars to train

ex-> GPT-4o is foundational Model of chatgpt.

Date _/_/_

Large language Models [LLM]

- Type of AI designed to generate coherent human-like text

- One notable example: GPT-4
- Trained on large corpus of text data
- Usually very big models
  - Billions of parameters
  - Trained on books, articles etc

Generative language Models

- We usually interact with LLM by giving a prompt.

- Then, model will leverage all the existing content it has learned from to generate new content.

- Non-deterministic ⇒ The generated text may be different for every user that uses the same prompt.

Date _/_/_

Amazon Bedrock

- Build Generative AI (GenAI) applications on AWS.
- Fully-managed service, no servers for you to manage
- Keep control of your data used to train the model
- Pay-per-use pricing model
- Unified APIs
- leverage a wide array of foundation models
- Out of box features: RAG, LLM Agents..
- Security, Privacy, Governance, and Responsible AI features.
- Access to wide range of foundational model [FM]
  ex: AI21labs, cohere, Stability.ai, amazon, anthropic, meta, mistral ai.
- Bedrock makes copy of the FM, available only to you, none of data is used to train FM.

Date _/_/_

Amazon bedrock - Base foundation Model

- How to choose?
  - Model types, performance requirements, capabilities, constraints, compliance
  - level of customization, model size, inference options, licensing agreements, context windows, latency.
  - Multimodal models

- What is Amazon titan?
  - High-performancing foundation models from AWS.

  - Image, text, multimodal model choices via a fully-managed APIs.
  - Can be customized with your own data.

- Smaller models are most cost effective.

Date _/_/_

Fine-tuning a Model

- Adapt a copy of a fountain model with your own data.
- fine-tuning will change the weights of the base foundation model.
- Training data must:
  - Adhere to a specific format
  - Be stored in Amazon S3.

Supervised fine-tuning

- Improves the performance of a model on specific tasks
- = further trained on a particular field or area of knowledge.
- Supervised fine-tuning uses labeled examples that are input-output pairs

Reinforcement fine-tuning

- Improves your FM model using feeback-based learning.
- You provide the input data [training data prompts]
- You define a Reward function to evaluate responses [output] and judge which responses are good.

Date _/_/_

Distillation

- Making models smaller and faster
- Up to 75% less expensive than original models
- decrease in accuracy, but potentially acceptable.
- Larger (teacher) model transfers knowledge to smaller (student) model
- Produces lighter model with similar behavior to the original.
- Focuses on efficiency, speed and cost reduction.

Note: - Re-training an FM requires a higher budget

- Supervised fine-Tuning is usually cheaper as computations are less intense and less data is usually required.

Date _/_/_

Note on Benchmark datasets

- Curated collections of data designed specifically at evaluating the performance of language models.
- Wide range of topics, complexities, linguistic phenomena.
- Helpful to measure: accuracy, speed and efficiency, scalability.
- Some benchmarks datasets allow you to very quickly detect any kind of bias and potential discrimination against a group of people.

Automated Metrics to evaluate an FM

- ROUGE:- Recall-Oriented Understudy for Gisting Evaluation.
  - Evaluating automatic summarization and machine translation systems
  - ROUGE-N: measure the no. of common n-grams b/w reference and generated text
  - ROUGE-L: longest common subsequence between reference and generated text.

Date _/_/_

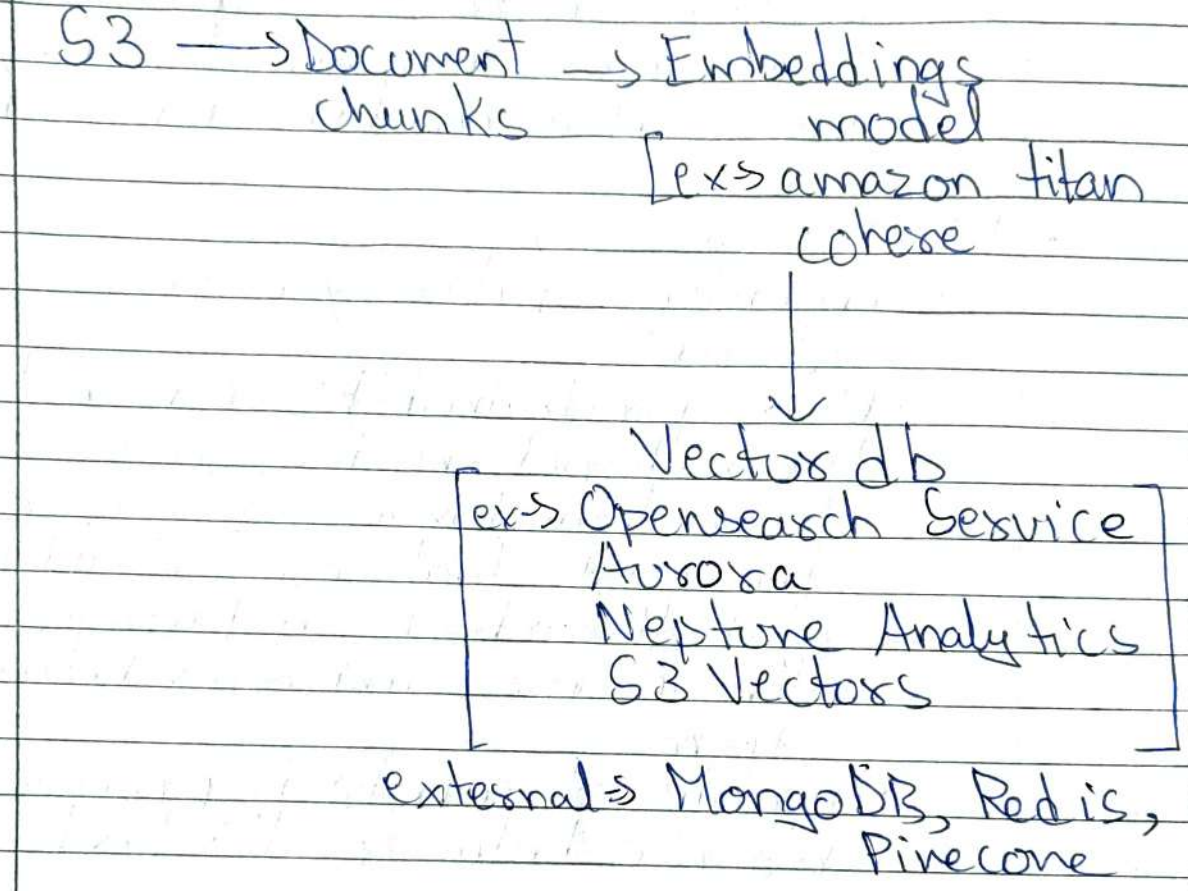
- BLEU: Bilingual Evaluation Udersstudy
  - Evaluate the quality of generated text, especially for translations
  - Considers both precision and penalizes too much brevity
  - looks at a combination of n-grams
- BERT Score
  - Semantic similarity between generated text.
  - Uses pre-trained BERT models [Bidirectional encoder representations from transformers] to compare the contextualized embeddings of both texts and computes the cosine similarity between them.
  - Capable of capturing more nuance between the texts

RAG & Knowledge base

- RAG = Retrieval-Augmented Generation
- Allows a foundational model to reference a data source outside its training data.
- Bedrock takes cares of creating Vector Embeddings in the db of your choice based on your data.

Date _ / _ / _

RAG: Vector Databases



★. Amazon OpenSearch Service [Serverless & Managed clusters]
Search & analytics db real time similarity queries, store millions of vector embeddings, scalable index management, fast nearest-neighbor (KNN) search capability

★. Amazon Aurora PostgreSQL
relational db, proprietary on AWS

Date _/_/_

- ★ Amazon Neptune Analytics
  graph db that enables high performance graph analytics and graph based RAGs (Graph RAGs)

- ★ Amazon S3 Vectors:-
  Cost effective durable storage with sub-second query performance.

GenAI Concepts - ~~Tokenizations~~

- Tokenization - converting raw text into a sequence of tokens
  - Word-based tokenizations: text is split into individual words
  - Subword tokenizations: Some words can be split too [helpful for long words]

- Context window
  - The number of tokens an LLM can considers when generating text.
  - The larger the context window, the more information and coherence
  - large context windows require more memory and processing power
- ★ First factor to look at when considering a model.

Date _ / _ / _

- Embeddings
  - Create vectors (array of numerical values) out of text, images or audio
  - Vectors have a high dimensionality to capture many features for one input token, such as semantic meaning, syntactic role, sentiment.
  - Embedding models can power search applications

* Words that have a Semantic relationship have Similar Embeddings

* - Guardrails
  - Control the interaction b/w users and foundation models [FMs]
  - filter undesirable and harmful content
  - Remove Personally Identifiable Information [PII]
  - Enhance privacy
  - Reduce hallucinations.
  - Ability to create multiple Guardrails and monitor and analyze user inputs that can violate the Guardrails.

Date _/_/_

Amazon Bedrock - Agents

- Manage and carry out various multi-step tasks related to infrastructure provisioning, application deployment, and operational activities
- Task coordination: perform tasks in the correct order and ensure information is passed correctly b/w tasks.
- Agents are configured to perform specific pre-defined action groups
- Integrate with other systems, services, databases and API to exchange data or initiate actions
- leverage RAGs to retrieve information when necessary.

Date _ / _ / _

Amazon Bedrock & Cloudwatch

- Model Invocation logging
  - Send logs of all invocations to Amazon CloudWatch and S3.
  - Can include text, images and embeddings
  - Analyze further and build alerting thanks to cloudwatch logs insights

- Cloudwatch metrics
  - Published metrics from Bedrock to cloudwatch
    - Including ContentFilteredCount, which helps to see if Guardrails are functioning
    - Can build Cloudwatch Alarms on top of Metrics

Amazon Bedrock - Pricing

- On-demand
  - Pay-as-you-go (no commitment)
  - Text models - charged for every input/output token processed
  - Embedding Models - charged for every input token processed
  - Image models - charged for every image generated

Date _/_/_

- Batch
  - Multiple predictions at a time (output in a single file in S3)
  - Can provide discounts of up to 50%
- Provisioned throughput
  - Purchase Model units for a certain time
  - Throughput - max number of input/output tokens processed per min
  - Works with Base, fine-tuned and Custom models

Model Improvement

$ I Prompt Engineering
- No model training needed [no additional computation or fine tuning]

$$ II Retrieval Augmented Generation (RAG)
- Uses external Knowledge
- No FM changes

$$$ III Instruction-based fine-tuning
- FM is fine-tuned with specific inst^n.

$$$$ IV Domain Adaptation fine-tuning
- Model is trained on a domain-specific dataset

Date _ / _ / _

Amazon Nova

- family of AI foundation Models(FMs) built by AWS.
- Think of it as AWS' alternative to chatgpt or claude
- Designed to be fast, cost-effective, enterprise-ready
- Understanding
  - ★ Nova Premier
  - ★ Nova Pro
  - ★ Nova Lite
  - ★ Nova Micro
- Creative
  - ★ Nova Canvas
  - ★ Nova Reel
- Speech → Nova Sonic

Date _/_/_

Prompt Engineering

- Prompt gives little guidance and leaves a lot to the model's interpretation
- Prompt Engineering = developing, designing, and optimizing prompts to enhance the output of FMs for your needs
- Improved Prompting technique consists of:
  - Instⁿ.
  - Context
  - Input data
  - Output Indicator

Negative Prompting

- A technique where you explicitly instruct the model on what not to include or do in its response.
- Negative Prompting helps to:
  - Avoid Unwanted Content
  - Maintain Focus
  - Enhance clarity

Date _/_/_

Prompt Performance Optimization

- System Prompts - how the model should behave and reply
- Temperature (0 to 1) - creativity of model's output
  - low (0.1) - outputs are more conservative
  - High (1.0) - outputs are more diverse
- Top P (0 to 1)
  - low P (0.2) - consider 25% most likely words.
  - High P (0.99) - consider a broad range of possible words.
- Top K - limits the number of probable words
  - low K (ex- 10) - more coherent response, less probable words
  - High K (ex 500) - more probable words, more diverse and creative
- length - max length of answers
- Max Sequences - tokens that signal the model to stop generating output.

Date _/_/_

Prompt latency

- latency is how fast model responds
- It is impated by :-
  - model size
  - mode type itself
  - no. of tokens in input
  - no. of tokens in output
- ★ latency is not impaced by topP, topK, temperature

Amazon Q Business

- fully managed Gen-AI assistant for your employees
- Based on your company's Knowledge and data
  - Answer questions, provide summaries, generate content, automate tasks
  - Perform routine actions
- Built on Amazon Bedrock (but you can't choose the underlying FM)

Date _ / _ / _

Amazon Q Business

- Data connectors [fully managed RAG] - connets to 40+ popular enterprise data sources
  eg → S3, RDS, Aurora etc
  gmail, Gdrive, etc
- Plugins: allows you to interact with 3rd party services
  - eg → Jira, ServiceNow, Salesforce etc
  - Custom Plugins
    Connect to any 3rd party application using APIs.

Amazon Q Business + IAM Identity Center

- Users can be authenticated through IAM Identity Center.
- Users receive responses generated only from the documents they have access to
- IAM Identity Center can be configured with external Identity Providers.

Date _/_/_

Admin Controls

- Controls and customize responses to your organizational needs
- Admin controls == Guardrails
- Block specific words or topics
- Respond only with internal information (vs using external knowledge)
- Global controls & topic-level controls

Amazon Q Apps

- Create Gen AI-powered apps without coding by using natural language
- leverages your company's internal data
- Possibility to leverage Plugins

Date _ / _ / _

Amazon Q for QuickSight

- Amazon Quicksight is used to visualize your data and create dashboards about them.
- Amazon Q understands natural language that you use to ask questions about your data.
- Create executive summaries of your data
- Ask and answer questions of data
- Grenerate and edit visuals for your dashboards.

Amazon Q for EC2

- EC2 instances are the virtual sesvers you can start in AWS.
- Amazon Q for EC2 provides guidance and suggestions for EC2 instance types that are best suited to your new workload.

Date _/_/_

Amazon Q for AWS chatbot

- AWS chatbot is a way for you to deploy an AWS chatbot in a slack or microsoft teams channel that knows about your AWS account.
- Troubleshoot issues, receive notifications for alarms, security findings, billing alerts, create support request

Amazon Q for Glue

- AWS Glue is an "ETL" [Extract transform and load] service used to move data across places

Bedrock

- GenAI app-building playground [powered by Bedrock]
- Allows you to experiment creating GenAI apps with various FMs (no coding or AWS accound required)

Date _/_/_

Q What is AI?
AI is a broad field for the development of intelligent systems capable of performing tasks that typically require human intelligence

AI Components

- Data layer - collect vast amount of data
- ML Framework and Algorithm layer
data scientists and engineers work together to understand use cases, requirements, and frameworks that can solve them
- Model layer - implement a model and train it, we have the structure, the parameters and functions, optimizer function

Q What is Machine learning (ML)?
- ML is a type of AI for building methods that allow machines to learn
- Data is leveraged to improve computer performance on a set of task.
- Make predictions based on data used to train the model.

Date _/_/_

Q What is deep learning (DL)?

- Uses neurons and synapses (like our brain) to train a model.
- Process more complex patterns in the data than traditional ML.
- Deep learning because there's more than one layer of learning.

Ex -> Computer vision, NLP

- large amount of input data, requires GPU

Neural Networks

- Nodes (tiny units) are connected together.
- Nodes are organized in layers
- When the neural network sees a lot of data, it identifies patterns and changes the connections b/w the nodes
- Nodes are "talking" to each other, by passing on (or not) data to next layer.

Date _/_/_

Q What is the transformer model? (LLM)

- Able to process a sentence as a whole instead of word by word.
- faster and more efficient text processing
- It gives relative importance to specific words in a sentence
- Transformer-based LLMs
  - Powerful models that can understand and generate human-like text.
  - Trained on vast amounts of text data from the internet, books, and other sources and learn patterns and relationship b/w words and phrases.

ML terms

- GPT (Generative Pre-trained transformer) generate texts or code based on input prompts
- BERT (Bidirectional Encoder Representations from transformers) similar to GPT but reads text in two directions

Date _/_/_

- RNN (Recurrent Neural Network)
  meant for sequential data such as time-series or text, useful in speech recongnition.
- ResNet (Residual Network)
  for image recognition
- SVM (Support vector Machine)
  ML algo for classification and regression
- WaveNet
  model to generate audio waveform
- GAN (Generative Adversarial Network)
  used to generate synthetic data such as images, videos or sounds.
- XGBoost (Extreme Gradient Boosting)
  implementation of gradient boosting

Date _/_/_

Supervised learning

- learn a mapping function that can predict the output for new unseen input data
- Need labeled data

Unsupervised learning

- The goal is to discover inherent patterns, structures or relationships within the input data.
- The machine must uncover and create the groups itself, but humans still put labels on the output groups
- Common techniques => Clustering, Association Rule learning, and Anomaly Detection

Semi-supervised learning

- Use a small amount of labeled data and a large amount of unlabeled data to train systems
- After that, the partially trained algo itself labels the unlabeled data called pseudo labeling.

Date _/_/_

- The model is then re-trained on result.

Self-supervised learning

- Having a model generate pseudo-labels for its own data without having humans label any data first
- Then, using the pseudo labels, solve problems traditionally solved by supervised learning

Reinforcement learning (RL)

- type of ML where an an agent learns to make decisions by perform-ing actions in an environment to maximize cumulative rewards.
- Key concepts
  - Agent - the learner or decision-maker
  - Environment - the external system the agent interacts with
  - Action - the choices made by the agent
  - Reward - feedback from environment
  - State - current situation
  - Policy - strategy uses to determine actions based on the state

Date _ / _ / _

RLHF (Reinforcement learning from Human feeback)

- Use human feeback to help ML models to self learn more efficiently

Model fit

- In case your model has poor performance, you need to look at its fit

- Overfitting
  - Performs well on training data
  - Doesn't perform well on evaluation data

- Underfitting
  - Model performs poorly on training data
  - Could be a problem of having a model too simple or poor data feature

- Balanced
  - Neither overfitting or underfitting
  - low bias, low variance

Date _/_/_

Bias

- Difference or error b/w predicted & actual value
- Occurs due to wrong choice in ML

→ High bias
- Considered as underfitting

→ Reducing the bias
- Use a more complex model
- Increase the number of features

Variance

- How much the performance of a model changes if trained on a different dataset which has a similar dist^{n}

→ High Variance
- Model is very sensitive to changes in training data
- This is overfitting

→ Reducing the variance
- less feature
- Split into training and test data set multiple times

Date _ / _ / _

Confusion Matrix

Actual Value	Predicted value: Positive	Predicted value: Negative
+ve	True Positive	False Negative
-ve	False Positive	True negative

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Date _/_/_

Metrics

- Precision - Best when false +ve are costly
- Recall - Best when false -ve are costly
- F1 score - Best when you want balance b/w precision and recall.
- Accuracy - Best for balanced datasets

AUC-ROC => Area under the curve - ~~rec~~ receiver opesator curve

Regression metrics

Accuracy

- MAE = Mean Absolute Error = $\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$
- MAPE = Mean absolute percentage error = $\frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$
- RMSE = Root mean squared error (RMSE) = $\sqrt{\sum_{i=1}^{n}\frac{(\hat{y}_i - y_i)^2}{n}}$

- R^2 => measure Variance

Date _/_/_

Inferencing

- Inferencing is when model is making prediction on new data
- Real time
  - Computers have to make decisions quickly as data arrives
  - Speed is prefered over perfect accuracy

  ex-> chatbot
- Batch
  - large amount of data that is analyzed all at once
  - Often used for data analysis
  - Speed of result is usually not a concern and accuracy is

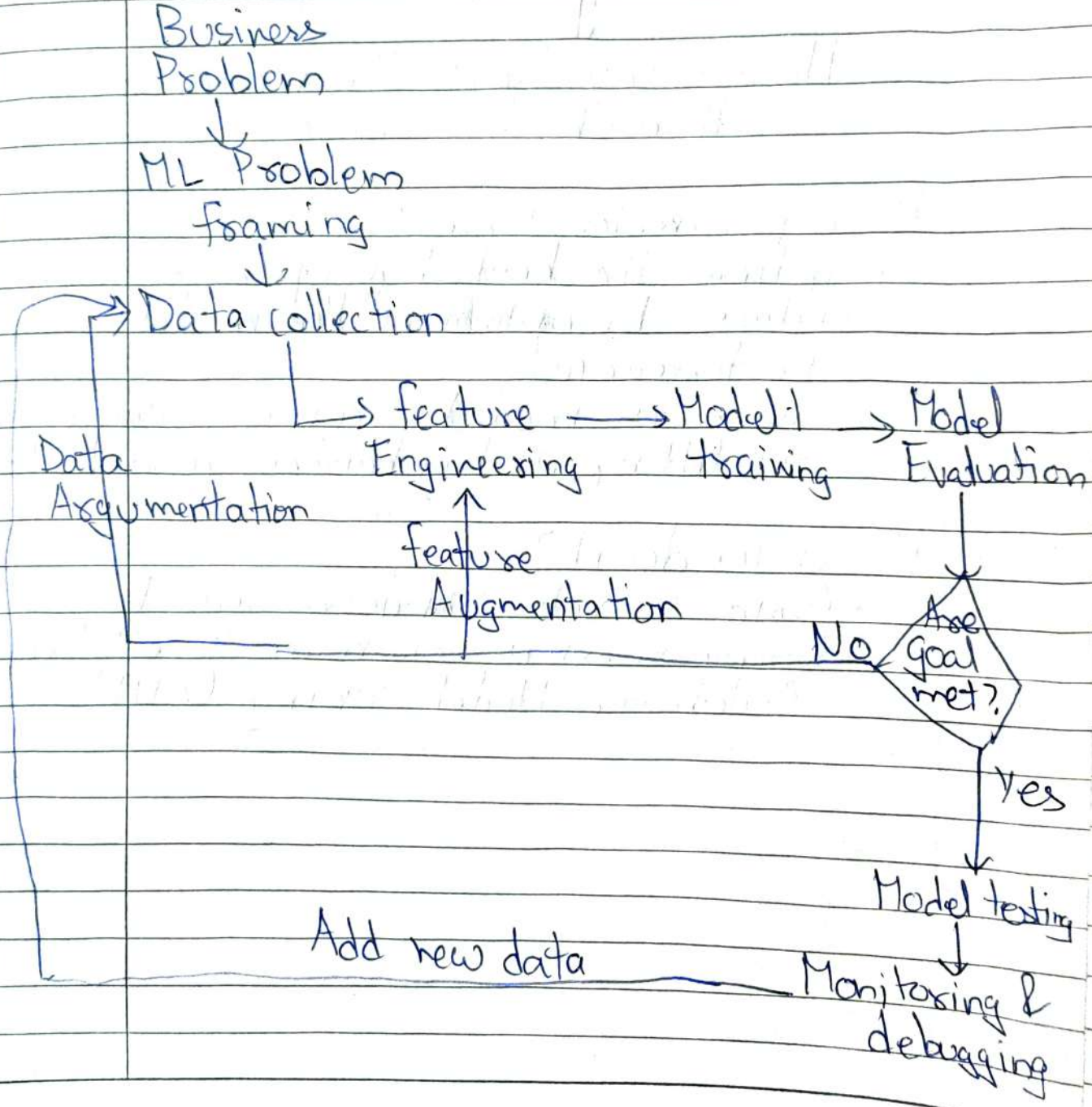
Inferencing at the Edge

- Edge devices are usually devices with less computing power that are close to where the data is generated,
- Small language model (SLM) on the edge device
  - very low latency
  - low compute footprint
  - Offline capability, local inference

Date _/_/_

- large language Model (LLM) on a remote servers
  - More powerful model
  - Higher latency
  - Must be online to be accessed

Phases of ML Project



Date _ / _ / _

Hyperparameter tuning

- Hyperparameter:
  - Settings that define the model structure and learning algo and process
  - Set before training begins
  - ex-> learning rate, batch size, no. of epochs, and regularization.
    - ↓
    - How many times the model works through that dataset.
- Hyperparameter tuning:
  - finding the best hyperparameters values to optimize the model performance
  - Improves model accuracy, reduces overfitting, and enhances generalization
- How to do it?
  - Grid search, random search
  - Using services such as SageMaker Automatic Model Tuning (AMT)

Date _/_/_

Q What to do if overfitting

- Overfitting is when model gives good predictions for training data but not for the new data.

- It occurs due to:
  - Training data size is too small and doesn't represent all possible input values
  - The model trains too long on a single sample set of data.
  - Model complexity is high and learns from the "noise" within the training data.

- How can you prevent overfitting?
  - Increase the training data size
  - Early stopping the training of model
  - Data augmentation (to increase diversity in the dataset)
  - Adjust hyperparameters (but you can't add them)
  - Ensembling (combine multiple models to get accurate results)

Date _/_/_

★ Amazon Comprehend

- for Natural language processing - (NLP)
- fully managed and serverless service
- Uses ML to find insights and relationship in text

- Custom Classification

- Organize documents into categories (class) that you define.
- Ex - categorize customer emails so that you can provide guidance based on type of customer request.

Named Entity Recognition (NER)

- Extracts predefined, general-purpose entities like people, places, organizations, dates, and other standard categories, from text.

Custom Entity Recognition

- Analyze text from specific terms and noun-based phrases
- Extract terms like policy no. or phrases that imply a customer escalation, anything specific to your business.

Date _/_/_

★ Amazon translate

- Natural and accurate language translation
- allows you to localize content - such as websites and applications - for international users, to easily translate large volume of text.

★ Amazon Transcribe

- Automatically convert speech to text
- Uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately
- Automatically remove Personally Identifiable Information (PII) using Redaction.
- Supports Automatic language Identification for multi-lingual audio

Allows transcribe to capture domain-specific or non-standard terms

- Custom Vocabularies
- Custom language models
- Toxicity detection

Date _/_/_

Amazon Polly

- Turn text into lifelike speech using deep learning
- Allowing you to create applications that talk.
- Lexicons
  - Define how to read certain specific pieces of text
  - Ex: AWS = "Amazon Web Services"
- SSML - Speech Synthesis Markup language
  - Markup for your text to indicate how to pronounce it
  - Ex: Hello, <break> how are you?
- Voice engine: generative, long-form, neural
- Speech mark
  - Encode where a sentence/word starts or ends in the audio
  - Helpful for lip syncing or highlighting words

Date _/_/_

Amazon Rekognition

- find objects, people, text, scenes in images and videos using ML.
- facial analysis and facial search to do user verification, people counting
- Create a db of "familiar faces" or compare against celebrities

→ Custom labels

- labels your training images and upload them to Amazon Rekognition
- Only needs a few hundred images or less.
- creates a custom model on images set

→ Content Moderation

- Automatically detect inappropriate, unwanted or offensive content.
- Integrated with Amazon Augmented AI (A2I) for human review

Date _/_/_

Amazon Lex

- Build chatbots quickly for your applications using voice and text.
- Supports multiple languages
- Integration with AWS Lambda, Connect, Comprehend, Kendra
- The bot automatically understands the user's intent to invoke correct lambda function to "fullfill the intent"

Amazon Personalize

- fully managed ML-service to build apps with real-time personalized recommendations
- Same technology used by Amazon.com
- Integrates into existing websites, applications, SMS etc.

→ Recipes
- Algo that base prepared for specific use.
- You must provide the training configuration on top of recipe.

Date _/_/_

Amazon textract

- Automatically extracts text, handwriting and data from any scanned documents using AI and ML

Amazon Kendra

- fully managed document search service powered by ML.
- Extract answers from within a document

Amazon Mechanical Turk

- Crowdsourcing marketplace to perform simple human tasks
- Distributed virtual workforce

Amazon Augmented AI (A2I)

- Human oversight of ML predictions in production

Date _/_/_

Amazon Transcribe Medical

- Automatically convert medical-related speech to text (HIPAA compliant)
- Ability to transcribes medical terminologies
- Supports both real-time (microphone) and batch (upload files) transcriptions

AWS HealthScribe

- HIPAA-eligible service that automatically generates clinical notes by analyzing patient-clinician conversations.
- Help you build clinical apps that provides rich transcripts, identifies speaker roles, classifies dialogues, extract medical terms, generate clinical notes.

Amazon EC2

- EC2 is one of the most popular of AWS offering
- EC2 = Elastic Compute Cloud = Infra as a Service

Date _/_/_

- It is mainly capable of
  - Renting virtual machines (EC2)
  - Storing data on virtual drives (EBS)
  - Distributing load across machines (ELB)
  - Scaling the services using auto scaling (ASG) group

Amazon SageMaker AI

- fully managed service for developers/ data scientists to build ML models
- Typically, difficult to do all the processes in one place + provision servers.

→ End-to-End ML Service

- Collect and prepare data
- Build and train ~~ml~~ ml models
- Deploy the models and monitor the performance of the predictions

Date _/_/_

- Built in Algorithms

- Supervised Algorithms
  - Linear regressions and classifications
  - KNN Algorithms (for classification)

- UnSupervised Algorithms
  - Principal Component Analysis (PCA) reduce no. of features
  - K-means - find grouping within data
  - Anomaly Detection

- Automatic Model tuning (AMT)

- Define the objective metric

- AMT automatically chooses hyper-parameter ranges, search strategy, maximum runtime of a tuning job and early stop condition

- Model Deployment & Inference

- Deploy with one click, automatic scaling no servers to manage (as opposed to self-hosted)

- Managed solution: reduced overhead

- Real-time, Serverless, Asynchronous

Date _/_/_

- Data Wrangler

- Prepare tabular and image data for ml.
- Data preparation, transformation and feature engineering
- Single interface for data selection, cleansing, exploration, visualization, and processing
- SQL Support, Data quality tool

Q What are ML features?

- features are inputs to ML models used during training and used for inference
- Ex -> music dataset

- Feature store

- Ingests features from a variety of sources
- Ability to define the transformation of data into feature from within feature store

Date _/_/_

Sagemaker Clarify

- Evaluate foundational models
- Evaluating human-factors such as humor
- leaverage an AWS-managed team or bring your own employees
- Use built-in datasets or bring your own dataset

Sagemaker Ground truth

- RLHF - Reinforcement Learning from Human Feedback
- Sagemaker Ground Truth Plus: Label data

SageMaker - ML Governance

- Sagemaker Model Cards
  - Essential model info
- Sagemaker Model dashboard
  - Centralized repository
- Sagemaker role Manager

Date _/_/_

SageMaker - Model monitor

- Monitor the quality of your model in production

SageMaker - Model registry

- Centralized repository allows you to track, manage and version ML models
- Catalog models, manage model versions, associate metadata with model

Sagemaker Pipelines

- A workflow that automates the process of building, training and deploying a ML model.
- Continuous Integration and continous Delivery (CI/CD) service for ML.
- Pipelines composed of steps and each step performs specific task.
- Supported Step types
  - Processing
  - Training
  - Tuning
  - AutoML
  - Model
  - Clarifycheck
  - Quality check

Date _ / _ / _

SageMaker Jumpstart

- ML Hub to find pre-trained foundational model.
- models can be fully customized for your data and use-case

SageMaker Canvas

- Build ML models using visual interface [no coding]
- Access to ready to use models Bedrock or JumpStart
- Build your own custom model using AutoML
- Leverage Data Wrangler for data prep.
- Have ready to use models

MLflow on Amazon SageMaker

- MLflow - an open-source tool which helps ML teams manage the entire ML lifecycle

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SageMaker Summary

- Sagemaker end to end ML service
- Sagemaker Automatic Model tuning [AMT]
  tune hyperparameters
- Sagemaker Deployment & Interence
  real time, serverless, batch, async
- Sagemaker studio
  unified interface for sagemaker
- Sagemaker Data Wrangler
  explore and prepare datasets, create features
- Sagemaker feature store
  store features metadata in a central place
- Sagemaker clarify
  compare models, explain model outputs, detect bias
- Sagemaker Ground truth
  RLHF, human for model grading & labeling data
- Sagemaker model Cards
  ML model documentation

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- Sagemaker Model Dashboard
  view all your models in one place
- Sagemaker Model Monitor
  monitoring and alerts for your model
- Sagemaker Model Registry
  centralized repository to manage ML model versions
- SageMaker Pipelines
  CI/CD for ML
- SageMaker Role Manager
  access control
- SageMaker JumpStart
  ML model hub & pre-built ML sol^n
- Sagemaker Canvas
  no-code interface for SageMaker
- MLflow on SageMaker
  use Mlflow tracking sesvess on AWS

Extra features

- Network isolation mode
- DeepAR forecasting algorithm

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Access token

digital ~~tok~~ Key, use in form of JWT

let's users access resources without needing to log in repeatedly.

- Users wants to access resources through client app.
- & users gives permission for the app to get an access token from authorization sesver.
- Sesver sends an access token to app.
- The app checks the token and, if valid, allows user to access resources.

very short lifespan.

Refresh token

-> Special token extends the life of an access token.

-> When access token expires, refresh token can be used to get a new one to get new one.
-> Authorization seaves usually stores token.
-> work togethes with access token to allow longes sessions.

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What is cloud computing?

→ The practice of using network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server.

On Premise	Cloud Providers
• You own the server	• Someone else owns the server
• You hire IT people	
• You pay or rent real state	
• You take all the risk	• We are just responsible for your configuring cloud services & code someone else takes care of the rest

Evolution of cloud Hosting

→ Dedicated Server
  1 physical machine ↔ to single business
• Very expensive, high maintenance, High security
→ VPS [Virtual Private Server]
  1 physical machine to single business
    ↳ is virtualized into sub-machines
runs multiple sites
• Better Utilization and Isolation of Resources

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→ Shared Hosting
1 physical machine, shared by hundred of businesses
- Very cheap, limited functionality, Poor isolation.

→ Cloud Hosting
Multiple physical machines act as one system
- Flexible, Scalable, Secure, Cost-effective, high configurability.

★ What is cloud Service Provider (CSP)?

It is a company which provides =>
- multiple cloud services
- Cloud services can be chained together to create architecture
- Cloud services are accessible via single unified API eg: AWS API.
- Cloud services utilized metered billing based on usage
- Cloud services have rich monitoring built in eg → AWS cloudtrail
- Cloud services have IaaS.
- Cloud services offers automation via IaC

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★ If a company offers multiple cloud services under a single UI but don't meet most of or all of these requirements it would be referred to as cloud platform
eg → Twilio, HashiCorp, Databricks. etc

Common cloud Services

There are 4 core services for IaaS

① Compute eg → EC2
② Networking eg → VPC
③ Storage eg → EBS
④ Databases eg → RDS

The Evolution of Computing

① dedicated

- Physical Server wholly utilized by a single customer.
- You have to guess your capacity
- you'll overpay for an underutilized Server.

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- You can't vertical scale, you need a manual migration.
- Replacing a server is very difficult
- You are limited by your Host operating System.
- Multiple apps can result in conflicts in resource sharing
- You have guarantee of security, privacy anf full utility of underlying resources.

(2) VMs

- You can run multiple VMs on one machine.
- Hypervisor is the software layer that lets you run the VMs.
- A physical server shared by multiple customers.
- You are to pay for a fraction of server.
- You'll overpay for an underutilized VM
- You are limited by your guest OS.
- Mutiple apps on a single VM can result in conflicts in resource sharing.
- Easy to export or import images for migration.
- Easy to ~~vertic~~ vertical or Horizontal Scale.

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(3) Containers

- VM running multiple containers
- Docker Deamon is the name of software layer that lets you run multiple containers.
- You can maximize the utilize of the available capacity which is more effective.
- Your containers share the same underlying OS so containers are more effcient than multiple VMs.
- Multiple apps can run side by side without being limited to the same OS requirements and will not cause conflicts during resource sharing.

(4) functions

- Are managed VMs running managed containers
- Known as Serverless Compute
- You upload a piece of code, choose the amount of memory and duration
- Only responsible for code and data, nothing else.
- Very cost-effective, only pay for the time code is running, VMs only run when there is code to be executed.
- Cold starts is side effect of this setup

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Types of Cloud Computing

① SaaS [for Customers]

A product that is run and managed by the service provider.

eg→ Salesforce, Gmail, etc

② PaaS

focus on the deployment and management of your apps.

Don't worry about, provisioning, configuring or understanding the hardware or OS.

③ IaaS

The basic building blocks for cloud IT. Provides access to networking features, computers and data storage space.

Don't worry about IT staff data centers and hardware.

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Cloud Computing Deployment models

1. Public Cloud

   Everything is built on the CSP

   also known as cloud first

2. Private cloud

   Everything built on company's datacenters also known as On Premise

   ex -> openstack

3. Hybrid

   Using both On-premise and CSP.

4. Cross-cloud

   Using Multiple cloud Providers aka multicloud

Amazon EKS <-> Azure Arc <-> GCP Kubernetes Engine

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→ Services like cloudfront & S3 are global doesn't need region.

~~EBS~~

Q What is Computing Power?

The throughput measured at which a computer can complete a computational task.

- General Computing
- GPU Computing
- Quantum computing

★ Amazon Braket → Quantum computing Service

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Benefits of cloud

- Agility
  -> Increase speed and agility
- Pay-as-you go pricing
  -> Trade capital expense for variable expense
- Economy of Scale
  -> Benefit from massive economies of scale
- Global Reach
  -> Go global in minutes
- Security
- Reliability
- High availability
- Scalability
- Elasticity

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7-advantages to Cloud

1. Cost-effective
2. Global
3. Secure
4. Reliable
5. Scalable
6. Elastic
7. Current

Q What is AWS Global Infrastructure?

It is globally distributed hardware and data centers that are physically networked together to act as one large resource for the end customers.

Regions are geographically distinct locations consisting of one or more Availability Zones

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Fault tolerance

A fault domain is a section of network that is vulnerable to damage if a critical device or system fails. The purpose of a fault domain is that if a failure occurs it will not cascade outside that domain, limiting the damage possible.

What is fault level?
A fault level is a collection of fault domains.

The scope of a fault domain could be
- specific servers in a rack
- an entire rack in a datacenter
- an entire room in a datacenter
- the entire data center building

It's upto CSP to define the boundaries of a domain.

Each Region is completely isolated from other regions.
=> This achieve the greatest possible fault tolerance and stability.

Each availability Zone [AZ] is isolated, but AZ in a region are connected through low latency links.

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Each AZ is independent failure zone.

Failure Zone

- AZ are physically separated within a typical metropolitan region and are located in lower risk flood plains.
- discrete uninterruptible power supply [UPS] and onsite backup generation facilities
- AZ are redundantly connected to multiple tier-1 transit providers.

- Multi AZ for High Availability

AWS Global Network

represent the interconnections b/w AWS Global infrastructure known as "Backbone of AWS".

Edge locations can act as on and off ramps to Global Network

AWS Global Accelerator
S3 transfer Acceleration
Uses edge locations as an on-ramp to quickly reach AWS resources in other regions by traversing the fast AWS Global Network

Cloudfront
Uses Edge locations as an off ramp, to provide at the Edge storage and compute near the end users.

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-> VPC Endpoints => ensuring your resources stay within the AWS Network and don't traverse over the public Network.

-> Points of Presence [PoP] is an intermediate location between an AWS Region and the end users, this location could be a datacenter or collection of hardware.

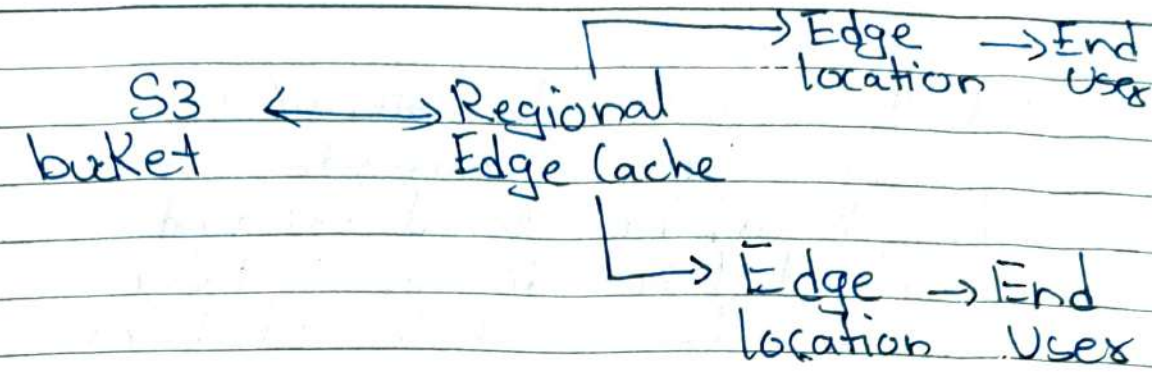
for AWS, PoP is a data center owned by AWS that is utilized related to content delivery or expediated upload.

PoP resources are => • Edge locations
• Regional Edge Caches

Edge locations are data centers that hold cached [copy] on the popular files so, that the delivery of distance to the end users are reduce.

Regional Edge locations are datacenters that hold much larger caches of less-popular files to reduce a full round trip and also to reduce the cost of transfer fees.

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Note → Tier 1 network is a network that can reach every other network on the Internet without purchasing IP transit or paying the peering.

AWS Availability Zones are all redundantly connected to multiple tier-1 transit providers.

The following AWS Services use PoPs for content delivery or expediated upload →

① Amazon Cloudfront is a content delivery Network [CDN]
- You point your website to cloudfront so that it will route requests to nearest Edge location cache
- allows you to choose an origin [such as web-server or storage] that will be source of cached
- caches the contents of what origin would returned to various E.L

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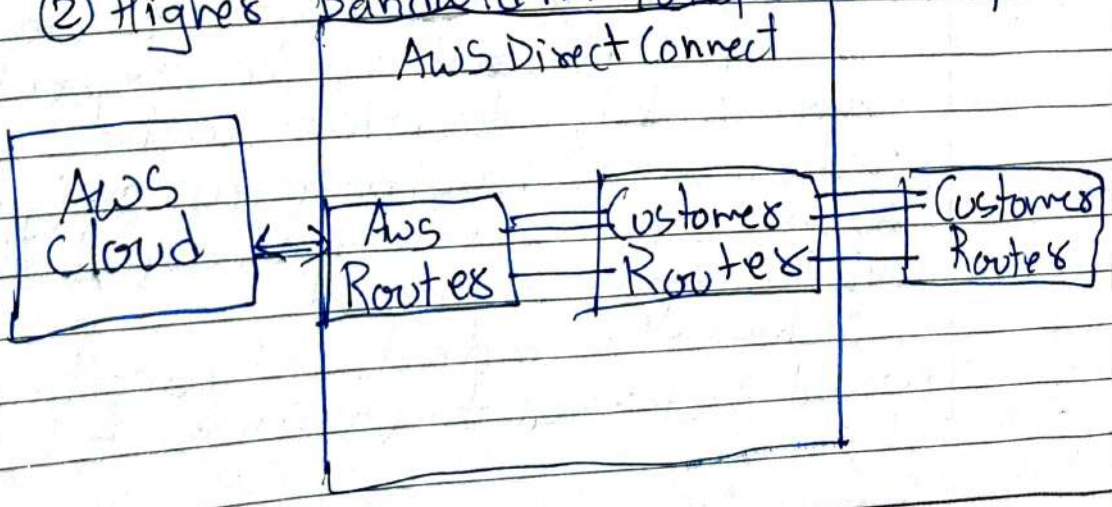
★ ② Amazon S3 transfer Acceleration allows you to generate special URL that can be used by end users to upload files to nearby Edge location. Once a file is uploaded to an Edge loc., it can move much faster within the Network.

★ ③ AWS Global Accelerator find the optimal path from end user to your web servers. Global Accelerator are deployed within edge locations so you send user traffic to an edge loc. instead to directly to your web-app.

AWS direct Connect

It is a private connection b/w your datacenter, office, co-location and AWS.

It has ① lower Bandwidth 50 MBps-500MBps
② Higher Bandwidth 1GBps or 10GBps.



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- Helps reduce network costs and increase bandwidth throughput.
  [Great for high traffic network]

- Provides a more consistent network experience than a typical internet-based connection. [reliable and secure].

→ Local Zones are datacenters located very close to a densely populated area to provide single-digit msec low latency performance [eg-> 7ms]

purpose is to support highly-demanding applications sensitive to latencies
- Media & Enterstainment
- Electronic Design Automation
- Ad-tech.
- Machine learning

→ AWS Wavelength Zones allows for edge-computing on 5g-networks. Applications will have ultra-low latency being as close as possible to the users.

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Q What is Data Residency?
Physical or geographic location of where an organization or cloud resources reside.

Q What is Compliance Boundaries?
A regulatory compliance [legal requirement] by a government or organization that describes where data and cloud resources are allowed to reside.

Q What is data Sovereignty?
data sovereignty is the jurisdictional control or legal authority that can be asserted over data because it's physical location is within jurisdictional boundaries.

★ -> AWS Outposts is physical rack of servers that you can put in your data center.
OR rack of servers running AWS Infra on your physical loc.

★ -> AWS Config is a Policy as Code service. You can create rules to continuous check AWS resources configuration. If they deviate from your expectations you are alerted or AWS config can in some cases auto-remediate

-> IAM Policies can be written explicitly deny access to specific AWS region.

-> A Service Control Policy (SCP) are permissions applied organization wide.

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Sustainabity

1. Renewable energy
2. Cloud efficiency
   AWS's infra is 3.6 times more energy efficient than median of U.S enterprise data centers.
3. Water Stewardship
   Direct evaporative technology to cool our data center Use of non-potable water for cooling purposes [recycled water]

☆ AWS Ground Station => is a fully managed service that lets you control satellite communications, process data, and scale your operations without having to worry about building or managing your own ground station infrastructure.

Usecase => 1. Weather forecasting
2. Surface imaging
3. Communications
4. Video broadcasts

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Q What is Solution Architect?
A role in a technical organization that architects a technical soln using multiple systems via researching, documentation, experimentation.

Q What is cloud Architect?
A solutions architect that is focused solely on architecting technical solutions using cloud services.

- Availability ⇒ Your ability to ensure service remains available.
- Scalability ⇒ Your ability to grow rapidly
- Elasticity ⇒ Your ability to shrink and grow to meet the demand.
- Fault tolerance ⇒ Your ability to prevent a failure
- Disaster Recovery ⇒ Your ability to recover from a failure [Highly Durable]

→ Elastic load Balancer ⇒ allows you to evenly distribute traffic to multiple servers in one or more datacenter. If a datacenter or server becomes unavailable the load balancer will route the traffic to only available datacenters with servers.

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High Scalability ⇒ ability to increase your capacity based on the increasing demand of traffic, memory and computing power.

Two types
- * Vertical Scaling ⇒ bigger servers [Scaling UP]
- Horizontal Scaling ⇒ Add more servers [Scaling Out]

High Elasticity ⇒ ability to automatically increase or decrease your capacity based on the current demand of traffic, memory and power.

Horizontal Scaling
- Scaling Out ⇒ Add more servers
- Scaling In ⇒ Removing underutilized servers

* Vertical Scaling is generally hard for traditional architecture. So, only horizontal Scaling described with elasticity.

⇒ Auto Scaling Groups [ASGr] ⇒ is an AWS feature that will automatically add or remove servers based on scaling rules you define based on metrics

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High fault tolerant ⇒ ability to ensure there is no single point of failure.

- fail overss is when you have a plan to shift traffic to a redundant system in case primary system fails

☆ → RDS Multi AZ → is when you run a duplicate standby database in another Availabil-ity Zone in case your primary database fails.

High dusability ⇒ ability to recover from disaster and to prevent loss of data

☆ → CloudEndure Disaster Recovery ⇒ continuously replicates your machines into a low-cost staging area in your target AWS account and preferred Region enabling fast and reliable recovery in case of data center failure.

Business Continuity Plan [BCP] ⇒ is a document that outlines how a business will continue operating during an unplanned disruption in services.

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Recovery Point Objective [RPO] => max acceptable amount of data loss after an unplanned data-loss incident, expressed as an amount of time

Recovery time Objective [RTO] => max amount of downtime your business can tolerate without incurring a significant financial loss.

AWS API

Q What is Application Programming Interface [API]?

An API is software that allows two applications/services to talk to each other.
eg => HTTP/S requests

→ AWS API is an HTTP API and you can interact by sending HTTPS requests. each service has its own Service Endpoint

Developer tools => (1) AWS Management Console & WISWIG Web Interface
(2) AWS SDK
Interact with API using your fav Programming lang
(3) AWS CLI
Interact with API via termind/shell

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Q What is powereshell? is a task automation and configuration management framework. A command-line shell and scripting language.

Amazon Resource Name (ARNs) uniquely identify AWS resources. ARNs are required to specify a resource unambiguously across all of AWS.

format variations =>
- arn:partition:service:region:account-id:resource-id

Q What is CLI? processes commands to a computer program in form of lines of text. OS implement a CLI in shell.

Q What is terminal? A terminal is text only interface [input/output environment]

Q What is Console? is a Physical computer to physically input information into a terminal.

Q What is shell? is a cmd line program that users interact with to input commands.
- Bash
- Zsh
- Powershell

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★ → AWS CLI allows users to programmatically interact with the AWS API via entering entering single or multi-line cmds into a shell or terminal.

→ AWS CLI is a Python executable program.
- Python is required to install AWS CLI
- ~~Free Aws~~

AWS SDK ⇒ is a collection of software development tools in one installable package

AWS CloudShell ⇒ is a browser-based shell ~~is~~ built into AWS Management Console.